

CME GLOBEX REFERENCE GUIDE

The CME Globex electronic platform brings diverse products to customers around-the-world nearly 24 hours a day.

As the world's leading and most diverse derivatives marketplace, CME Group (www.cmegroup.com) is where the world comes to manage risk. CME Group exchanges offer the widest range of global benchmark products across all major asset classes, including futures and options based on interest rates, equity indexes, foreign exchange, energy, agricultural commodities, metals, weather and real estate. CME Group brings buyers and sellers together through its CME Globex electronic trading platform and its trading facilities in New York and Chicago. CME Group also operates CME Clearing, one of the largest central counterparty clearing services in the world, which provides clearing and settlement services for exchange-traded contracts, as well as for over-the-counter derivatives transactions through CME ClearPort. These products and services ensure that businesses everywhere can substantially mitigate counterparty credit risk in both listed and over-the-counter derivatives markets.

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ELECTRONIC TRADING AT CME GROUP

GLOBAL ACCESS TO ALL YOUR MAJOR ASSET CLASSES FROM A SINGLE PLATFORM

The CME Globex platform was the first—and remains among the fastest—global electronic trading system for futures and options. Through its advanced functionality, high-reliability and global connectivity, it is now the world's premier marketplace for derivatives trading. CME Globex gives you access to the broadest array of futures and options products available on any exchange, virtually around the clock, from anywhere in the world.

CME Group electronic markets are among the most liquid in the world, enabling customers to choose from a wide range of products in thousands of possible expirations and spreads. In addition to now having a single point of connection, customers who trade CME, CBOT, NYMEX and COMEX futures and options enjoy streamlined development efforts, a single source for customer support and more efficient access to real-time market data.

The platform is continuously enhanced to serve customers with high-speed, high-volume capacity, improved options capabilities and a range of new products. Functionality and capacity continue to grow dramatically to accommodate demand.

Benefits include:

Performance and speed

Trades on CME Globex are executed in less than the blink of an eye and instantly confirmed back to the customer anywhere in the world. CME Group is constantly working to reduce processing time—the platform continues to handle significantly more volume at a significantly faster pace. Our average daily order volume continues to increase, while the response time to execute a trade continues to decrease.

Diverse, innovative products

CME Globex is the first derivatives platform to offer global access to all major asset classes—interest rates, equity indexes, FX, agricultural commodities, energy, metals, weather, real estate and environmental commodities. Customers can trade a range of global benchmark futures and options, including our flagship Eurodollar and E-mini S&P 500 contracts, benchmark metals and energy products, and highly innovative alternative investment products such as weather and real estate futures. Further, through partnerships with exchanges like BM&FBOVESPA, Bursa Malaysia, the Dubai Mercantile Exchange, the Minneapolis Grain

Exchange, the Kansas City Board of Trade, and the Green Exchange, CME Globex gives you simplified access to market exposure around the world.

Open access and direct participation

The platform is an open access marketplace that allows customers to participate directly in the trading process, view the book of orders and prices for CME Group products and enter their own orders. Anyone who has an account with a Futures Commission Merchant (FCM) or Introducing Broker (IB), who in turn has a CME Clearing guarantee, can trade on the platform.

Access the CME Globex platform from nearly 100 countries and foreign territories through a range of connectivity options.

Our electronic trading platform offers customers:

- Streamlined development efforts
- Single source for production customer support
- Access to free risk management tools
- More efficient access to real-time market data
- 10-deep futures and 3-deep options for most markets
- High-speed, high-volume capacity
- Improved options capabilities and a range of new products
- Direct order routing to Brazilian exchange BM&FBOVESPA
- Access to listed products traded on the Dubai Mercantile Exchange (DME), the Kansas City Board of Trade (KCBT), the Minneapolis Grain Exchange (MGEX), Bursa Malaysia (BMD) and the Green Exchange

Connectivity Options

Customers worldwide have access to the platform from nearly 100 countries and foreign territories, with new connections being added all the time. The range of connectivity options include Internet connections, direct connections and several telecommunication hubs that provide reduced connectivity costs, increased accessibility and fast efficient trading. Our telecommunication hubs are located in Amsterdam, Dublin, Kuala Lumpur, London, Paris, Milan, São Paulo, Seoul and Singapore.

Fairness, transparency, anonymity

CME Group electronic markets provide an entirely level playing field for all participants. Individual retail traders, small businesses and large institutions alike see and have access to the same prices. The anonymity of traders and firms is protected electronically in all bids, offers and execution reports.

Market integrity

CME Clearing, the largest derivatives clearing house in the world, guarantees all contracts traded on the CME Globex platform and ensures that all traders honor their performance obligations. The CME Globex Control Center (GCC) monitors trading activity and system performance around-the-clock to assure the orderly functioning of CME Group electronic markets.

Learn more about the connectivity options and benefits delivered by the CME Globex platform at www.cmegroup.com/globex.

PRODUCT OVERVIEW – WHAT'S ACCESSIBLE ON CME GLOBEX

The CME Globex platform brings diverse product solutions to customers around-the-world nearly 24 hours a day. These products address the business and risk management objectives of customers which are as varied as our products. All electronically-traded CME Group products are fully fungible (interchangeable) with their CME Group floor-traded equivalent contracts. Product codes, price and data streams, however, may be distinct for the electronic and floor-listed contracts.

To view a complete list of CME Group products available on the CME Globex platform as well as contract codes, up-to-date specs and expiration dates for all CME Group products, visit www.cmegroup.com.

Click on the product names below to go to a description for that product.

- [CME Group Interest Rate Products](#)
- [CME Group Equity Indexes](#)
- [CME Group FX Products](#)
- [CME Group Agricultural Products](#)
- [CME Group Real Estate Products](#)
- [CME Group Weather Products](#)
- [CME Group Metals Products](#)
- [CME Group Energy Products](#)

CME Group Interest Rate Products

CME Group Interest Rate products span the entire U.S. dollar-denominated yield curve. Customers can use our products to manage short-, medium- and long-term interest rate risk with products based on Eurodollars, U.S. Treasuries, LIBOR, OIS and Fed Funds, and interest rate swaps. These products provide trading opportunities and resources for risk management related to interest rate exposures ranging from overnight to 30-year investment horizons.

Our Eurodollar futures are the world's most actively traded interest rate contracts and a benchmark for global investors, and our options on Eurodollar futures comprise the largest and most liquid options markets in the world. Our benchmark U.S. Treasury contracts are the world's most widely traded contracts for measuring long-term investment risk.

View our interest rate contracts and specifications at www.cmegroup.com/ir.

CME Group Equity Indexes

CME Group is the leading equity index derivatives marketplace, offering futures and options on key benchmark indexes that cover the spectrum of small-, mid- and large-cap indexes in the United States, Europe and Asia. Our global product suite includes contracts on such well-known indexes as the S&P 500, S&P MidCap 400,

S&P SmallCap 600, Dow Jones Industrial Average, NASDAQ-100, Nikkei 225 Stock Average, S&P CNX Nifty and more.

View our equity contracts and specifications at www.cmegroup.com/equities.

CME Group FX Products

CME Group is the largest regulated marketplace for FX (foreign exchange) trading, and the second largest electronic marketplace for FX trading in the world. We offer 52 futures contracts and 31 options contracts based on 20 global currencies, including major world currencies and currencies of emerging markets.

Our FX volume has grown significantly in recent years, with an average daily notional value of trades that exceeds \$100 billion. Our FX market serves as an effective and efficient means of risk transfer for the global FX market, bringing together a broad array of client segments—including banks, hedge funds, proprietary trading firms and active individual traders—by offering investment as well as risk-management opportunities.

View our FX contracts and specifications at www.cmegroup.com/fx.

CME Group Agricultural Products

CME Group offers the widest range of agricultural commodity futures and options available on any exchange. Our agricultural contracts include grains, oilseeds, livestock, dairy products, lumber, coffee, sugar, cocoa and other products. These products are the staples of everyday life and represent the origins of the earliest forward and futures markets. We serve agricultural producers and those seeking risk management and hedging tools, alongside funds and other traders looking to capitalize on the extraordinary opportunities our markets offer.

View our agricultural contracts and specifications at www.cmegroup.com/commodities.

CME Group Real Estate Products

The CME Group Real Estate marketplace covers both U.S. residential and commercial properties. These products can be used by a broad range of market participants, from property owners and manufacturers to hedge funds and investors.

View our real estate contracts and specifications at www.cmegroup.com/realestate.

CME Group Weather Products

Weather futures are another example of the long-standing CME Group tradition of innovation. One-third of businesses worldwide are directly affected by weather conditions. Our weather contracts enable companies to manage weather-related risk while also offering opportunities to speculate — absorbing that risk in exchange for possible profit on weather variations. Our weather product suite offers trading opportunities related to temperatures, snowfall, frost and hurricanes. The products are based on a range of weather conditions in more than 47 cities in the United States, Europe, Canada, Australia and Asia, with the hurricane products geared to nine U.S. regions.

View our weather contracts and specifications at www.cmegroup.com/weather.

CME Group Metals Products

Incorporating NYMEX and COMEX futures and options, our Metals markets include full-size contracts on gold, silver, platinum, palladium, copper and steel; and smaller size contracts for gold (miNY 50 oz.), (E-micro 10 oz.), silver (miNY 2,500 oz.) and copper (E-mini 12,500 lbs.). With an average daily volume of more than 400,000 futures and options contracts traded, our Metals markets are the most liquid in the world for these

products. Additionally, we offer clearing services of OTC London Gold spot and forwards and Iron Ore Swap futures through CME ClearPort.

View our metals contracts and specifications at www.cmegroup.com/metals.

CME Group Energy Products

CME Group offers the most extensive and liquid energy complex in the world, including Light Sweet Crude Oil (WTI), natural gas (Henry Hub), petroleum and electricity products. Many of our contracts are benchmarks that set the price for these resources worldwide. From the world's largest industrial companies to financial institutions, our diverse universe of participants clear an average daily volume of 1.6 million energy contracts every day on CME Globex, through CME ClearPort or on our trading floor. Building on the legacy of NYMEX, CME Group is the world's leading energy marketplace, and is growing stronger every day.

View our energy contracts and specifications at www.cmegroup.com/energy.

ACCESSING THE CME GLOBEX PLATFORM

AN OPEN ARCHITECTURE THAT EASILY ACCOMMODATES CHANGING MARKETS, NEW CUSTOMERS AND NEW PRODUCTS

The open architecture of the CME Globex platform provides a wide range of access solutions and extensive functionality.

The platform readily accommodates changing markets, new customers and new products. In addition to the trade matching system, components of the platform include trading applications, an order entry interface, a market data dissemination system and order management tools. These components all work together seamlessly to provide virtually instantaneous market visibility and trading.

A Wide Range of Trading Solutions

Traders may develop a proprietary front-end application or select from a variety of certified applications to execute electronic orders and receive market data directly. These applications are available from CME Group, FCMs, IBs and Independent Software Vendors (ISVs), and vary in the types of functionality supported.

On the CME Group trading floors, traders can access CME Group electronic markets through the GALAX-C hand-held electronic trading system and through third-party hand-held solutions. The Denali system is designed to provide customers and FCMs who trade on both CME Globex and our trading floors the functionality to route orders to the CME Group trading floors.

Learn more about front-end trading solutions at www.cmegroup.com/tradingapplications.

The CME Globex Access Directory is a guide to the front-end solutions available for trading on the CME Globex platform. View the current CME Globex Access Directory at www.cmegroup.com/globexaccessdirectory.

CME Group Solutions The CME Globex Order Routing Interface: iLink

iLink provides direct access to all CME Globex markets and supports all the enhanced trading functionality available on the platform. Based on the industry-standard FIX 4X protocol, it readily supports customized trading systems.

Learn more about iLink at www.cmegroup.com/ilink.

Market Data Platform and FIX/FAST Market Data Format

The Market Data Platform is a dual-feed multicast market data dissemination system that allows customers to integrate CME Group market data directly into their trading applications. It offers superior scalability and efficient delivery of market data messages with no middleware requirements.

The FIX/FAST market data format, greatly improves bandwidth scalability and provides for reduced latency. FIX/FAST is the basis of a new industry standard for market data. The new format is based on FIX and FAST protocols for increased efficiency and disseminates market data for all CME Group markets and Partner Exchanges.

For more on FIX/FAST and Market Data Platform, visit www.cmegroup.com/fixfast.

CME EOS Trader

Customers can access advanced options functionality of the CME Globex platform through CME EOS Trader, a CME Group-provided trading application that connects to CME Globex over a secure Internet connection. With CME EOS Trader, customers can trade any CME Globex options products with virtually 24-hour access to CME Globex markets.

Benefits Include:

- Request for Quote (RFQ) and Request for Cross (RFC) functionality
- Strategy Builder Window feature that enables quick and intuitive creation of complex strategies
 - Offers ability to quickly create custom User Defined Strategies (UDS)
 - Simplifies creation of Delta Neutral or “Covered” strategies

For further information please visit www.cmegroup.com/eos.

SMART Click

The SMART Click tool is a secure web-based interface that can be used by CME Globex customers to create a SMART Click user ID and manage your profile information, including contact details. Currently, SMART Click user IDs can be used in conjunction with the following CME Group systems:

- FirmSoft 6.0
- CME Globex Credit Controls (GC2)
- Clearing Firm Administrator Dashboard

As more CME Group systems become integrated with SMART Click, users will benefit from the following features of the SMART Click platform:

- Access to all SMART Click-integrated systems will be available with one user ID and password;
- A single SMART Click user ID can support access to CME Group systems with permissions for multiple clearing firms when appropriate;
- The SMART Click user retains the SMART Click user ID even when changing clearing firms or employers; and,
- Self-service updates to the SMART Click profile allow the user to make changes to profile information with no paperwork required.

CME Globex Risk Management Tool

Cancel on Disconnect

Cancel on Disconnect (COD) monitors the iLink system for involuntary lost connections between iLink users and the CME Globex platform. If a lost connection is detected, COD cancels all resting futures and options orders for the disconnected registered iLink user. This does not include GTC (Good 'til Cancel) and GTD (Good 'til Date) orders. It is the user's responsibility to reenter all orders that have been canceled by COD. There is no charge for the COD service.

Features:

- Opt-in, subscription-based
- Free service

Learn more about this tool at <http://www.cmegroup.com/cod>.

CME Globex Credit Controls (GC2)

CME Globex Credit Controls (GC2) provides pre-execution risk controls and allows Clearing Firm Risk Administrators to set real-time credit limits on all CME Globex customers. Clearing Firm Risk Administrators can define trading limits and also select real-time actions if those limits are exceeded. GC2 is intended to complement existing Clearing Firm risk systems and front-end risk tools. Credit values are calculated using the margin rate of reference multiplied by the contract size of top day positions and resting orders for all Globex executed exchanges. GC2 access is only available at the Class A Clearing Member level. As stated in the CME mandatory Credit Control Policy (Rule 949), clearing members must employ GC2 credit control functionality on every one of their CME Globex Executing Firms.

Learn more about this tool at www.cmegroup.com/gc2.

FirmSoft

FirmSoft is a free, browser-based order management tool that provides real-time visibility to working and filled orders across multiple firm IDs. FirmSoft provides important alternative access to working and filled orders in case of system failure. FirmSoft provides view and cancel capabilities for all CME Globex orders.

Users can view:

- Order information, including status
- Fill information, including partial fills
- History
- Details, including timestamps and cancel replace history

Users can also cancel an individual order, a group of orders or all working orders for which they are permissioned with FirmSoft.

Learn more about this tool at www.cmegroup.com/firmsoft.

THE TRADING SESSION

Trading sessions on the CME Globex platform meet customer needs around-the-world.

Electronic Trading Availability

Virtually all CME Group products trade electronically at some point during the day, but specific schedules vary according to product.

Note: Special holiday trading hours are available at www.cmegroup.com/holiday.

A current list of products on CME Globex and specific availability can be found online at www.cmegroup.com/trading_hours.

Trading Sessions and States

Trading on the CME Globex platform is generally available Sunday evening through late Friday afternoon. Exact trading hours vary by product. CME Globex sessions start in the afternoon or evening and mark the beginning of the next trading day (for example, orders entered during Sunday's evening session are dated for and cleared on the following Monday).

Market States

Most CME Globex markets cycle through five daily order entry states. During each trading session, all futures contracts transition through these predefined states:

1. Market Enabled/Pre-Opening

A predetermined time before the trading session opens when customers can begin entering, modifying and canceling orders for the next trading day, but no trades are executed.

2. Pre-Opening/No-Cancel

A predetermined time before the session opens when traders can only enter orders for the next trade date but cannot cancel, execute or modify orders, but no trades are executed.

3. Continuous Trading

The period of time when orders are sent and matched in real time, based on the product's trading times.

4. Surveillance Intervention

The period of time that begins when CME Globex sends the notification that trading has ended for a group of products for that particular trade date. During this time, only order cancellation can occur.

5. Post Close/Pre Open (PCP)

This market state allows order placement, modification, and cancellation of GTC/GTD orders only. No matching takes place and no action can be taken on non GTC/GTD orders. This is only available for the Grain markets and occurs between 13:15 - 13:30 Central Time (CT) and 14:30 CT and 16:00 CT.

6. Maintenance Period

The period of time Monday through Thursday when trading is closed on the CME Globex platform for regularly scheduled maintenance.

The ability to enter, modify or cancel orders varies according to the CME Globex predefined state, as the chart below shows:

	Order Entry	Order Modification	Order Cancellation
CME Globex Session Starts 1. Market Enabled/Pre-Opening	✓	✓	✓
2. Pre-Opening/No Cancel	✓		
MARKET OPEN			
3. Continuous Trading	✓	✓	✓
4. Post Close/Pre-Open (PCP)*	✓	✓	✓
MARKET CLOSE			
5. Surveillance Intervention			✓
CME Globex Session Ends 6. Maintenance Period			

■ No order functionality available

Note: All open session orders are automatically canceled at market close.

* GTC and GTD orders only.

ENTERING ORDERS ON CME GLOBEX

The CME Globex platform supports a broad array of order functionality, offering convenience and flexibility to meet a wide variety of individual trading needs. This section summarizes the

order types and order duration qualifiers available in the CME Group markets.

Supported Order Types and Functionality on CME Globex – By Product Line

Order Types	Agriculture	Equities	FX	Interest Rates	Energy	Metals	Real Estate	Weather
Limit	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Market with Protection	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Market to Limit	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Stop Limit	F	F	F	F	F	F	F	F
Stop with Protection	F	F	F	F	F	F	F	F
Minimum Quantity	F, O	F, O	F, O	F	F, O	F	F	F
Hidden Quantity	F, O	F, O	F, O	F	F, O	F, O	F	F

Functionality	Agriculture	Equities	FX	Interest Rates	Energy	Metals	Real Estate	Weather
Cross Order	O	O	O	O	O	O		
Mass Quote	O	O	O	O	O	O		
UDS: Covereds	O	O	O	O	O	O		
UDS: Combos	O	O	O	O	O	O		

KEY: F = Available for futures O = Available for options

* Some CME Group Interest Rate Products support the RFC functionality. Please see the iLink and FIX/FAST at www.cmegroup.com/globexsdk for more information.

Order Types

The availability of specific order types varies based on how customers access CME Globex and the products they trade, as the chart above shows.

Limit

A Limit order allows the buyer to define the maximum price to pay and the seller the minimum price to accept (the limit price). A Limit order remains on the book until the order is either executed, canceled or expires.

Market with Protection

Market orders at CME Group are implemented using a “Market with Protection” approach. Unlike a conventional Market order, where customers are at risk of having their orders filled at extreme prices, Market with Protection orders are filled within a predefined range of prices (the protected range). The protected range is typically the current best bid or offer, plus or minus 50 percent of the product’s Non-Reviewable Range. If any part of the order cannot be filled within the protected range, the unfilled quantity remains on the book as a Limit order at the limit of the protected range.

Stop Limit

A resting Stop Limit order is triggered when the trigger price is traded in the market. The order then becomes a Limit order with the customer's specified limit price. The order is executed at all price levels between the trigger price and the limit price. If the order is not fully filled the unfilled quantity remains in the market at the limit price. A buy Stop Limit order must have a trigger price greater than the last traded price for the instrument. A sell Stop Limit order must have a trigger price lower than the last traded price for the instrument.

Stop with Protection

A Stop with Protection is a Stop Limit order (see above) with the limit price calculated based on:

- The trigger price, and
- The protected range.

The protected range is typically the trigger price, plus or minus 50 percent of the Non-Reviewable Range for that product. The limit price for a buy Stop with Protection will be calculated by adding the protected range to the trigger price. Likewise, the limit price for a sell Stop with Protection will be calculated by subtracting the protected range from the trigger price. Once the limit price for the order is calculated, it becomes a Stop with limit order in all respects.

Minimum Quantity

A Minimum Quantity order is immediately executed only if a certain minimum quantity of that order can be immediately matched. If the minimum quantity cannot be immediately filled, the entire order is canceled.

Hidden Quantity

A Hidden Quantity order—also called MaxShow or Iceberg—displays only a portion of the order to the marketplace. When the displayed quantity has been filled, another portion equal to the displaced quantity is then displayed.

Request for Cross (RFC)

A Cross order is a pre-negotiated trade between two parties in which one party (preferably the seller) enters the buy and sell sides in a single message, with the same price and quantity for the buy and sell sides. Both sides of the Cross order are exposed to the market and may be matched against new and resting orders from all market participants. The remaining quantities on the buy and sell sides are then matched to each other. For rules related to Cross order process and regulation, please refer to [CME Rule 533](#).

Mass Quote

A Mass Quote is a bundle of two-sided, actionable quotes in one or more options markets submitted via a single message. Only designated customers can submit Mass Quotes, and they have special protections that help mitigate their risk in matched transactions.

Additional Notes on Order Types

- Some order types require an opposing bid or offer to be present in the market before the order can be submitted. Entering one of these order types without an opposing bid or offer present will cause the order to be rejected.
- Front-end system vendors may offer additional functionality in their applications. Check with your vendor for more information.

Order Duration Qualifiers

Order duration qualifiers allow you to indicate how long the order will be valid – for example, if it should remain open for the next trading session if it isn't executed by the end of the trading session in which it was entered. An order submitted without an order duration qualifier will be eliminated at the end of that same trading session if not filled, or if it was received between trading sessions, at the end of the following session.

As with order types, not all order duration qualifiers are available for all CME Group products. The following table provides a quick reference to the order duration qualifiers supported by product line.

Day

An order to buy or sell a contract during that trading day only. Day orders that have been placed but not executed on a

trade date do not carry over to the next trade date. Unexecuted Day orders are eliminated at the end of the day.

Good 'Til Canceled (GTC)

GTC orders will remain in force until executed, canceled or the contract expires.

Good 'Til Date (GTD)

GTD orders remain in force through the end of the specified date unless executed or canceled, or the contract expires.

Fill and Kill (FAK)

FAK orders are immediately filled in whole or in part at the specified price or better. Any remaining quantity is eliminated.

Fill or Kill (FOK)

FOK orders are canceled if not immediately filled for the specified minimum quantity at the specified price or better.

Additional Functionality

UDS

User-Defined Spreads (UDS) functionality enables customers to create and trade unique instruments intra-day. A UDS: Covered instrument is composed of an outright option or options strategy covered by one or more underlying outright futures.

A UDS: Combos instrument is composed of any or all of the following:

- Exchange-defined outright option(s)
- Existing user-defined spread(s)

Order Duration Qualifiers – By Product Line

Duration Qualifier	Agriculture	Equities	FX	Interest Rates	Energy	Metals	Real Estate	Weather
Session/Day	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Good 'Til Canceled (GTC)	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Good 'Til Date (GTD)	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Fill and Kill (FAK)	F, O	F, O	F, O	F, O	F, O	F, O	F	F
Fill or Kill (FOK)	F, O	F, O	F, O	F, O	F, O	F, O	F	F

KEY: F = Available for futures
O = Available for options

MARKET PROTECTIONS

The following features minimize excessive price movements on the CME Globex platform and ensure fair, competitive markets.

Stop Spike Logic

Stop Spike Logic prevents the excessive price movements caused by cascading stop orders by introducing a momentary pause in matching (Reserved State) when triggered stops would cause the market to trade outside predefined values (typically the same as the Non-Reviewable Ranges). This momentary pause allows new orders to be entered and matched against the triggered stops in an algorithm similar to market opening.

Whenever a lead month futures instrument is placed in a Reserved State, the Options Auto-Reserve functionality automatically pauses matching in the associated options markets. All resting Mass Quotes are canceled when the Auto-Reserve functionality is initiated. This state is maintained for a few seconds after the futures contract has resumed trading. During the reserved period, customers can submit, modify and cancel all orders except Market Orders and Mass Quotes; Mass Quotes and Market Orders are rejected.

Price Banding

To ensure fair and orderly markets, CME Group has a price banding mechanism in place that subjects all incoming electronic orders to price verification and rejects all orders with clearly erroneous prices. Price bands are monitored throughout the day by the CME Globex Control Center (GCC) and adjusted if necessary. A dynamic enhanced options price banding system is implemented for options and options spreads traded on the CME Globex platform.

For complete banding information on each CME Globex product, including information on the enhanced options price banding, visit

www.cmegroup.com/pricebanding.

Reserve Price Band Multiplier

When a non-implied Lead Month and/or Secondary Month futures instrument transitions into the Reserved State, the price band values are expanded by a predefined Reserve Price Band Multiplier. The Reserve Price Band Multiplier expands the default Price Band by a positive whole number multiplier (e.g., 2x or 3x) during the Reserved State, to allow Limit and Stop order entry from the non-biased Buy or Sell side of the market. Allowing Limit and Stop order entry supports the creation of an Indicative Opening Price (IOP) at a fair market value level, thus supporting a more accurate reopening.

Matching Algorithms

To ensure that customers get the best possible executions at the fairest prices, the CME Globex platform employs predefined sets of matching rules—algorithms—designed to best meet the needs of market participants in each product group. View details on the algorithms used for the different product groups at

www.cmegroup.com/algorithm.

MARKET OPERATIONS

The CME Globex Control Center (GCC)

The GCC provides market operations, support and customer service for all electronic trading on CME Globex. That includes handling all CME Globex inquiries and requests for assistance from CME Group-registered contacts. The GCC supports the matching engine, order routing and market data interfaces and network connectivity.

Additionally, the GCC also provides customers with order status and, in cases of emergency, may cancel orders for customers. The GCC will only provide critical assistance to CME Group registered contacts. Critical assistance consists of order status, order cancellation and net position. If you are not registered, have your firm administrator call the GCC on your behalf. To become a CME Group-registered contact, customers must complete [Schedule 3 & 4 - User ID and GCC Authorized Contacts Request](#).

Areas of expertise include:

- Product and market information
- Block trade reporting
- Performance troubleshooting
- GCC notifications
- Product scheduling
- Trading halt management
- Limit move procedures
- Error trade oversight and resolution
- Liability claim support
- Order status and cancellation
- Settlement and limit price maintenance
- Price banding oversight
- Product state management (launching, listing, delisting, activations and expirations)
- CME EOS Trader functionality

Hours

The GCC is available from 2:00 p.m. CT Sunday through 4:45 p.m. CT Friday. An international number is provided for the convenience of our global customers.

How to contact the GCC

U.S.:

+1 312 456 2391

Europe:

+44 20 7623 4708

Asia:

+65 6223 1357

Order Cancel/Status Requests

In order to process an order cancellation or status, the GCC requires CME Group-registered contacts to be able to confirm their identity (through security information) and provide:

1. User ID/Trader ID
2. Nature of inquiry
3. Trade Order Number (TON)
4. Account number
5. Instrument, buy or sell and quantity

GCC Phone Requests – Quick Navigation Guide

This quick reference guide to the automated GCC telephone system is designed to help customers navigate the phone system more quickly

Note: You do not need to wait for the menu prompts to play, you can select your option at any time.

CME GLOBEX CONTROL CENTER PHONE MENU

GCC – +1 312 456 2391

GCC Outside U.S. – +44 20 7623 4708 / +65 6223 1357

OPTION	DESCRIPTION:	SELECT FOR:
1	Invoke the Error Trade Policy or Report a Block Trade	Invoke the Error Trade Policy or Report a Block Trade
2	Order Status, Order Cancellation	Critical Order Status Assistance
3	Order Routing, Market Data, or CME web based application support	Assistance for issues relating to Production Connectivity, Order Entry, Market Data, General Functionality, CME EOS Trader and FirmSoft Support
4	CME Globex Products and Markets	All other requests for market assistance for production trading (e.g., product specifications, holiday calendars)

Note: You can select the desired option at any time after the menu begins.

For GCC Administrative Assistance:	+1 312 580 5387
For Certification and New Release Environments, call CSET:	+1 312 930 2322
CME ClearPort Facilitation Desk:	+1 800 438 8618 / +1 212 299 2670

All calls to the GCC are recorded. www.cmegroup.com/gcc

GCC Notification System

To keep customers informed, especially during system emergencies, the GCC also provides CME Group-registered customer contacts with real-time CME Globex updates via telephone, e-mail and pager. In emergency situations, special procedures are implemented to enhance information sharing.

If CME Group determines that an event is likely to affect a significant number of customers or may require an emergency market halt, the information is disseminated via GCC Targeted Messaging, Market Data Platform administration messages, the CME Group Web site and/or the GCC Phone Menu.

Registered contacts also can sign up for GCC Targeted Messaging to receive messages via e-mail, pagers, mobile phones and land phones.

If markets are halted, the GCC will send regularly timed messages throughout the investigation of the event. Once the issue has been resolved, the GCC will send appropriate messages with the same methods used to announce that the event has been resolved.

In the event of a Market Halt, the GCC will also make this information available to all customers via www.cmegroup.com.

Trade Cancellation Policy

If an order execution occurs at a price not representative of the current market, the CME Group customer may call the GCC to request to have the trade reviewed.

Some important notes:

The customer must call the GCC within eight minutes of the error trade. A trade stands if the GCC is not notified within the eight-minute limit. The GCC also sends out a message (including the contract, time of trade, quantity and trade price) to all CME Globex registered contacts via Targeted Messaging alerting them that the trade has been questioned.

In order to cancel or price adjust a trade, the GCC must first determine if the trade is within the Non-Reviewable Range:

- If a trade is within the Non-Reviewable Range, the trade will stand and cannot be canceled or price adjusted.
- If a trade is outside the Non-Reviewable Range, the GCC will send a message (including the contract, time of trade, quantity and trade price) to all CME Globex registered contacts via Targeted Messaging and the quote vendor network alerting them that the trade has been canceled, adjusted or stands. The GCC will also attempt to contact directly all parties involved in the trade.

Note: All futures and options contracts are price adjusted, not canceled, except in certain circumstances. View more information on the trade cancellation and price adjustment procedures in [Rule 588](#), including the Non-Reviewable Ranges for each product.

CME Globex Messaging Policy

The CME Globex Messaging Policy creates fair business guidelines by which customers are billed a surcharge for overly high message rates. This policy benefits all customers trading on CME Globex by discouraging excessive messaging abuses, which in turn helps to ensure that CME Globex maintains the responsiveness and reliability that our customers around-the-world have come to expect from it when trading the CME Group electronic markets. Under the CME Globex Messaging Policy, each clearing member firm (active or inactive clearing member firms that maintain relationships with CME Clearing) must not exceed product-specific benchmarks, individually tailored to the valid trading strategies of each market.

CME Group calculates benchmarks based on a per-product Volume Ratio, defined as the number of messages submitted for each executed contract in a given product. CME Group compares the total number of messages to traded volume for all markets from 7 a.m. – 4 p.m. CT with the exception of Equities, which occurs from 7 a.m. – 3:15 p.m. CT, then applies a variation percentage.

If a clearing member firm exceeds a benchmark, they will be issued two notices within a rolling 30 business day period. Subsequently, a \$2,000 surcharge, per product, per session, per clearing member firm, will be billed when a benchmark is exceeded. CME Group-designated market makers are assigned different benchmarks. Clearing member firms may request reconsideration of surcharges within 10 business days of the event.

More information on the CME Globex Messaging Policy and the current product benchmarks is available at www.cmegroup.com/messagingpolicy.

The Liability Claim Process

Customers who take market action based upon an incorrect order status provided by the GCC or a CME Group system, service or facility may complete a Liability Claim Form to seek damages. In general, CME Group may accept liability for direct losses incurred by a customer who takes market action based upon an incorrect order status provided by the GCC or a CME Group

system, service or facility. CME Group liability in such situations is subject to cumulative liability caps. A customer who receives an incorrect status must take immediate action to limit any losses as soon as the customer knows or should have known that the order status information is incorrect.

In addition, CME Group will not be liable if it provides prior notification that a CME Group system, service or facility may produce such incorrect order status information and a means to access the correct information. CME Group liability to such a customer will be limited to the price at which the customer's position could have reasonably been liquidated.

If you have any questions regarding the Liability Claim Process, please contact the GCC or refer to the complete text of the policy in the Liability Claim Form document posted on the CME Group Web site.

CME Clearing

In today's market environment, effective risk management is a business imperative. Risk management opens the door to opportunity and it supports faster and smarter decisions. CME Group's centralized clearing capability brings the benefits of transparency, trust and confidence to our markets. By acting as the counterparty to every trade, CME Clearing mitigates counterparty credit risk, meaning that for every trade completed on our exchange, we guarantee the financial soundness of both parties. By being at the center of every transaction, we can protect everyone's interest and ensure the integrity of the marketplace.

Key Features of CME Clearing:

- **Your counterparty credit risk is substantially mitigated** as CME Clearing becomes the buyer to every seller and the seller to every buyer. We guarantee the financial soundness of both parties.
- **Accumulation of debts/losses is limited** and transparency is achieved as CME Clearing marks your positions to market twice daily.

- Approximately \$8 billion in financial safeguards stand ready to mitigate systemic risk.* In over a century of excellence, CME Clearing has never had a default.
- Customer segregation protections under which your funds are not subject to creditor claims against your own clearing firm should it become financially unstable or insolvent are included.
- Multilateral netting helps you gain capital efficiencies and decreases your credit exposure.
- Your choice of execution is preserved and made safer. You have the flexibility to manage your risk on-exchange or with cleared-only services for OTC products, all with the performance guarantee of CME Clearing.

Electronic Market Data (E-data) Information

CME Group provides customers with access to real-time, delayed and historical market data via Internet-based E-data applications.

Real-Time Data

CME E-quotes

The CME E-quotes application offers a best-in-class display of data. In both web-based desktop and mobile editions, CME E-quotes delivers real-time, delayed and historical quotes and charts, and analytics to your exact specifications. Benefit from global access and a powerful combination of sophistication, usability, performance and price.

For more information about CME E-quotes, visit www.cmegroup.com/e-quotes.

CME DataSuite

CME DataSuite lets you see over 200 CME, CBOT and NYMEX futures and options markets in action with just a click of your mouse.

- Free, real-time quotes are listed within 1-second accuracy
- See futures quotes with five levels of depth
- Options quotes listed with one level of depth

To access CME DataSuite, visit www.cmegroup.com/datasuite.

Delayed Data

Charts

This application provides current and historical performance of CME Group contracts with free delayed snapshot charting.

Delayed Futures and Options

Customers can get detailed delayed price information at each product and contract level.

Historical Data

CME DataMine

With a broad array of data types, CME DataMine makes it easier than ever to access historical price information in a form that can be easily used to develop, enhance or refine trading algorithms. By providing market depth, top-of-book, time and sales, and end-of-day data, CME DataMine has established itself as a premier information resource. Customers can access daily updates through a secure FTP account as well as conduct one-time bulk orders.

For more information on CME DataMine, visit www.cmegroup.com/datamine.

* Approximately \$8B in financial safeguards as of September 30, 2010.

Important Contacts for Electronic Traders

Contact Information	Department	Phone
CME Globex Development and Connectivity	CME Global Account Management U.S. Europe Asia	+1 312 634 8700 +44 20 7796 7100 +65 6593 5574
Market and System Inquiries	CME Globex Control Center (U.S.) U.S. Europe Asia	+1 312 456 2391 +44 20 7623 4708 +65 6223 1357
Product Information	CME Group Customer Service	+1 800 331 3332

CME GLOBEX ELECTRONIC TRADING

RULES AND POLICIES

The following list of rules and policies is not meant to be exhaustive or complete, but is intended as a summary of the most common CME Globex rule or policy questions. For the text of actual rules or interpretations, please refer to the CME, CBOT or NYMEX Rulebooks, which are available online at www.cmegroup.com, or the various Market Regulation Advisory Notices or other policies that have been published on the Web site. This document should not be relied upon as the official source of the rules. Further, while many rules have been harmonized across CME Group Exchanges, several rules remain different and market participants must be aware of these differences. Specific differences are noted with respect to the rules referenced below.

Accessing CME Globex

Rule 574 outlines certain requirements for gaining access to CME Globex, including a requirement that all connections to CME Globex be guaranteed by a clearing member firm. Additionally, all individuals entering orders in other than a clerical capacity on behalf of non-member customers must have appropriate industry registration. These orders may be entered only from designated areas on the trading floors or from the premises of an entity registered to conduct customer business.

Any clearing member firms providing CME Globex access to their customers must comply with all Credit Control requirements as set forth in Schedule 6 of the Customer Connection Agreement.

Market Data Usage on CME Globex

Any entity connecting to CME Group for market data and/or distributing market data is required to sign the Market Data License Agreement (MDLA) and complete a Scope of Use document. The Market Data License Agreement (MDLA) covers acceptable uses of CME Globex market data used for trading purposes for CME Group and participating Partner Exchanges.

There are certain criteria that must be met to qualify for a CME Globex market data fee waiver. The main three criteria are that a user/device must only be accessing market data for electronically traded products from CME Globex, the user/device is capable of routing orders to CME Globex, and the user has a capitalized futures account capable of trading CME Group's or Participating Exchange's products. Failure to meet any of these criteria will require execution of the Market Data License Agreement (MDLA) and any payment of applicable fees.

In addition, the mandatory reporting requirements associated with CME Globex market data usage are defined in the Market Data License Agreement (MDLA). Reporting is due monthly and all firms are responsible for reporting data usage that they directly control.

For additional information regarding the Market Data License Agreement (MDLA), visit www.cmegroup.com/marketdatapolicy.

CME Globex Trade Cancellations and Price Adjustments

Rule 588 states that trade prices determined by GCC to be within the Non-Reviewable Range will generally not be canceled or adjusted. The only exception to this rule is when the GCC determines that allowing a trade to stand as executed may have a material, adverse effect on the integrity of the market.

The GCC may review a trade based on its analysis of market conditions or a request for review by a CME Globex user. A request for review must be made as soon as possible, but will generally not be considered if more than eight minutes have passed since the trade occurred.

A party responsible for an order(s) which results in a trade price adjustment or a trade cancellation may be liable for realized losses incurred by affected parties pursuant to the criteria outlined in **Rule 588**.

The Non-Reviewable Ranges for all products and the Bid/Ask Reasonability Allowance for options trading on CME Globex are set forth in **Rule 588**.

Identification of CME Globex Terminal Operators

Rule 576 requires that CME Globex terminal operators be identified to the Exchange in the manner prescribed by the Exchange, and are subject to Exchange rules. For parties accessing CME Globex through iLink, clearing member firms or ISVs will issue the workstation user ID (known as the Tag 50 ID).

The user ID must be submitted with each order sent to CME Globex. Additionally, some user IDs are required to be registered with the applicable Exchange by an Exchange clearing member firm. In no event may a member or clearing member firm enter or permit another party to enter an order with a user ID other than the individual's own unique user ID.

Clearing member firms must maintain historical records of the identities of each user for at least five years. Additionally, please see the most recent Market Regulation Advisory Notice concerning Tag 50 IDs and registration requirements which may be accessed here www.cmegroup.com/rulebook/rulebook-harmonization.html

Rules and Requirements for Entering Orders on CME Globex

This section highlights the most commonly referenced rules and requirements for entry of orders into CME Globex. **It is not intended as an exhaustive list of all such rules and requirements.**

- Every order entered into CME Globex must include an account designation, although there is an exception to the requirement of entering the actual account number if the order is entered by an eligible account manager for accounts eligible for post-execution allocation as set forth in **Rule 536.C**.
- Pursuant to **Rule 530**, no person may enter an order into CME Globex for an account they own, an account in which they have a financial interest or an account over which they have discretionary trading authority (including an order allowing discretion as to time and price) when such person is in possession of any order for another party that CME Globex is capable of accepting.
- Pursuant to **Rule 548**, all non-discretionary customer orders must be entered into CME Globex in the order received. If the non-discretionary order is not able to be entered immediately upon receipt, it must be entered when it becomes executable.
- Pursuant to **Rule 529**, no person may withhold or withdraw any order from the market for the benefit of any person other than the person placing the order.
- Pursuant to **Rule 531**, a person in possession of a customer order may knowingly trade against his customer order for his own account only if the customer order has been entered immediately upon receipt and has first been exposed on CME Globex for a minimum of five seconds in the case of futures orders or for a minimum of 15 seconds in the case of options orders.

- Pursuant to **Rule 532**, no person may disclose another person's order or solicit or induce another person to disclose order information. An exception to this prohibition is set forth in each Exchange's **Rule 539.C**, which allows parties to engage in pre-execution communications regarding CME Globex trades in certain products provided that the parties comply with the requirements set forth in the rule. Please see the section on pre-execution communications for a more detailed discussion of the rules regarding pre-execution communications concerning CME Globex trades.
- Pursuant to **Rule 534**, no person shall place or accept buy and sell orders in the same product and expiration month, and, for a put or call option, the same strike price, where the person knows or reasonably should know that the purpose of the orders is to avoid taking a bona fide market position exposed to market risk.

Automated Trading Systems

An Automated Trading System (ATS) is an electronic system or computer software that is programmed to generate and send orders into CME Globex in an automated or semi-automated fashion. An ATS controller is a person who operates and/or administers the ATS. Typically, this is a person who initiates or disables particular algorithms or strategies, or adjusts the parameters of the automated program.

CME Globex Record Keeping and Records Retention Requirements

For each order entered into CME Globex, **Rule 536.B** requires the terminal operator to input the price, quantity, product, expiration month, Customer Type Indicator (CTI code) and account number, and, for options, put or call and strike price. Definitions concerning each of the four CTI codes are set forth in **Rule 536.D**.

Each order entered into CME Globex must be entered with the correct workstation operator identifier as described earlier in this document.

Pursuant to **Rule 536.B**, terminal operators who receive orders that are capable of and are immediately entered into CME Globex are not required to create a written, time-stamped order. Orders that cannot be entered immediately into CME Globex must be reduced to writing and include the date, time stamp, order instructions and account designation. CME Globex users are encouraged to print and save a listing of all orders they enter into CME Globex during each session.

Clearing member firms are responsible for maintaining or causing to be maintained for five years an audit trail for electronic orders entered into CME Globex through iLink, and must be able to display this data in a pre-approved format using terminology and field names consistent with other CME Group systems. Please see **Rule 536** in each Exchange's Rulebook for more information on record keeping requirements.

Pre-Execution Communications Concerning CME Globex Trades

In certain markets, market participants are allowed to engage in pre-execution communications with regard to transactions executed on CME Globex in circumstances where one party wishes to be assured that a contra party will take the opposite side of the order. Such pre-execution communications are allowed in all CME and NYMEX futures and options and in options on CBOT Interest Rate, Ethanol and Dow futures. **Pre-execution communications are prohibited in all CBOT futures products and in CBOT agricultural options.** In order for parties to engage in pre-execution communications, they must adhere to the requirements and procedures set forth in each Exchange's **Rule 539.C**.

Some circumstances in which pre-execution discussions might take place include the following:

- A firm may wish to contact potential contra parties on behalf of its customers or for the benefit of the firm proprietary account, in order to gain some assurance that someone will be ready and willing to take the opposite side of the order. These opposite parties could be the firm's customers or house traders, other firms' customers or proprietary traders, or individual traders.
- An individual member may also contact potential contra parties to trade opposite his or her personal trading account.

More detailed information, including the text of CME **Rule 539.C**, and a question and answer document concerning the policies and procedures concerning CME Globex pre-execution discussions are available on the CME Group Web site via the following link:

www.cmegroup.com/rulebook/rulebook-harmonization.html

Pre-execution communications concerning trades intended for open outcry/pit execution are expressly prohibited and constitute a violation of **Rule 539**.

APPENDIX – GLOSSARY

After-Hours Electronic Contracts (i.e., ETH only) After-hours electronic contracts trade on the CME Globex platform only after they stop trading via open outcry on the trading floor.

Application Program Interface An application program interface (API) is a set of tools and protocols that help integrate a solution with other software programs or facilitate the customization of a user interface.

CME Clearing The division of CME Group that confirms, clears and settles all trades. CME Clearing also collects and maintains performance bond funds, regulates delivery and reports trading data.

CME Connect The portal entry point for CME Globex customers on the Web site.

CME Globex The CME Group electronic trading platform.

CME Globex Control Center (GCC) The department that supports and maintains the CME Group electronic trading environment.

Continuous Trading Phase The phase on the CME Globex platform in which orders are sent to the market for real-time matching.

Cross Margin The process of allowing for a reduction in performance bond (margin) requirements. This reduction is possible because risk is reduced when offsetting positions are cleared by the same or affiliated clearing members.

CSET See *Customer Support for Electronic Trading*.

Customer Forum A communication vehicle to update customers and solicit feedback on topics related to CME Group's electronic trading strategy over the next six to 12 months, including new policies, enhancements and products.

Customer Support for Electronic Trading (CSET) The department that assists with certification and development to CME Globex.

Denali System that couples broker-to-broker trade matching with automated trade checking and immediate match status notification directly to the traders in the trading floor. This is accomplished through the legacy CBOT order routing and electronic card ("HHT") applications and the newly developed trade matching engine.

Derivatives Financial instruments whose value is based upon other financial instruments, such as stock indexes, interest rates or interest rate indexes.

Electronic-Only Contracts Electronic-only contracts trade exclusively on the CME Globex platform.

E-px (e-"packs") A Web-based application that provides real-time Eurodollar pack and bundle data, including prices for the 40 underlying quarterly Eurodollar futures contracts and a daily log of all executed Eurodollar spread trades.

E-quotes A CME Group real-time market data application that offers streaming quotes directly from CME Group for all futures products, in addition to interactive charts and OsterDowJones news.

Error Trade A trade that falls outside a contract's Non-Reviewable Range and is deemed to have been submitted in error. Such trades can be removed from official CME Group records, including Time and Sales information.

EOS Trader A Web-based application provided by CME Group that supports enhanced electronic options trading functionality, including User-Defined Spreads, as well as some limited futures trading functionality.

FCM See *Futures Commission Merchant*.

FEC-ETP See *Front-End Clearing System Electronic Trade Posting*.

FIX/FAST The new market data format offered by CME Group that is based on FIX and FAST protocols and designed to provide increased efficiency.

FirmSoft A user-friendly, browser-based order management business continuity tool offered at no cost by CME Group to provide firms easy access to view and cancel their orders.

Front-End Clearing System Electronic Trade Posting (FEC-ETP) Application that enables brokers to inquire about CME Globex net positions and spread prices. FEC-ETP is accessible from any PC with an Internet connection.

Fungibility Interchangeability of contracts. For example, five E-mini S&P 500 contracts are fungible with one standard-sized S&P 500 contract.

Futures Commission Merchants (FCM) Individuals, associations, partnerships, corporations and trusts that solicit or accept orders for the execution of a commodity transaction on and pursuant to the rules of a contract market, and which accept payment from or extend credit to customers.

GCC See *CME Globex Control Center*.

GCC Notification System The method the GCC uses to send messages regarding system events to customers via e-mail, pagers, fax, mobile phones and land phones.

Hand-held The GALAX-C electronic hand-held trading system enables local traders to access CME Globex products directly from the open outcry trading floors.

Hedging Offsetting price risk in a cash market by taking an equal but opposite position in the futures market. A means of protection against loss due to adverse price fluctuations.

Hidden Quantity A Hidden Quantity order – also called Max Show or Iceberg – displays only a small portion of the order to the marketplace. When the displayed quantity has been filled, another portion is then displayed.

IB See *Introducing Broker*.

iLink The order routing interface to CME Globex, based on the FIX 4.2 protocol. iLink creates a session between CME Group and the client through FIX protocol, eliminating the need for Standard Template Libraries (STLs).

Implied Functionality Implied prices integrate bids and offers in both spreads and their outright contracts to provide the most liquid possible markets with the best possible prices.

Independent Software Vendor (ISV) A company that makes and sells software products that run on one or more computer hardware or operating systems.

Indicative Opening Price (IOP) The price at which a product would trade upon the open, if possible, or the opening bid or offer if no trade would occur. At predefined times prior to the opening on CME Globex, an IOP is calculated as orders are entered for each product. Trading begins at an equilibrium price that falls within the overlap of the bid and offer prices. The IOP is broadcast to all CME Globex users and to the CME Group market data/ticker feed.

Internet Service Provider (ISP) A company that provides Internet access.

Introducing Broker (IB) A firm or person engaged in soliciting or accepting and handling orders for the purchase or sale of futures contracts, subject to the rules of a futures exchange, but not accepting any money or securities to margin any resulting trades or contracts. The IB is associated with a correspondent futures commission merchant and must be licensed by the Commodity Futures Trading Commission (CFTC).

ISP See *Internet Service Provider*.

ISV See *Independent Software Vendor*.

Local Pre-Processing Additional functionality an ISV may provide to its customers before the order is transmitted to the CME Group electronic trading platform.

Maintenance Period A period of time in each CME Globex trading session in which the market remains closed and customers cannot enter, modify, view or cancel orders. All remaining day, session and expired Good 'Till Date (GTD) orders are purged during this time.

Market Data Platform A dual-feed multicast market data dissemination system used in conjunction with the CME Group order entry interface. Market Data Platform offers superior scalability and more efficient data delivery.

Market Enabled/Pre-Opening Period The period in the CME Globex trading day in which the market is enabled and customers can begin to enter orders.

Minimum Quantity A Minimum Quantity order is executed only if a certain minimum quantity of that order can be immediately matched.

Non-Reviewable Range A range of prices used in determining if a potential error trade should be canceled. The range is based on the true market price for the contract immediately before the error trade occurred, as determined by considering all relevant information, including the last trade price on the CME Globex platform, a better bid or offer price, a more recent price in a different contract month, the price of the same or a related contract established in open outcry trading and the prices of related contracts trading in other markets (e.g., cash FX and SGX Eurodollars). A trade may not be canceled if it falls within the Non-Reviewable Range for that contract. Non-Reviewable Ranges vary by product.

Order Duration Qualifiers Order Duration Qualifiers define the span of time the order will be active. Default order duration is Day/Session.

Post Close/Pre-Open Phase The phase in the CME Globex trading day in which customers are allowed to enter orders, modify and cancel GTC and GTD on the grain markets only.

Pre-Opening/No-Cancel Phase The phase in the CME Globex trading day in which customers are allowed to enter orders, but are prohibited from modifications and cancellations.

Price Adjustment To alter the price of an executed trade because the order was entered or filled in error and fell outside the Non-Reviewable Range (a price range) for that product. Trades are adjusted according to strictly followed established rules in the CME Globex Error Trade Policy.

Price Banding A CME Group-instituted mechanism to ensure a fair and orderly market. This mechanism subjects all incoming orders to price verification and rejects all orders with clearly erroneous prices. Price bands are monitored throughout the day by the GCC and adjusted if necessary.

Protection Points Used to define the protected range of Market and Stop orders with Protection. The protected range is typically the current best bid or offer, plus or minus 50 percent of the Non-Reviewable Range for that instrument.

Side-by-Side Contracts Side-by-Side Contracts trade simultaneously on the CME Globex platform and on the trading floor via open outcry for a portion of the day.

Software Development Kits (SDKs) A set of documents providing info on the message specifications, architectures and functionality to facilitate development to CME Globex.

Surveillance Intervention Phase A period of time in each CME Globex trading session during which the market is closed and customers are not allowed order entry or modification, although cancellation of orders is permitted. The Surveillance Intervention Period is followed by the CME Globex Maintenance Period.

Trade Cancellation To remove a trade from the official CME Group records because the order was entered or filled in error and fell outside the Non-Reviewable range (a price range) for that product. Trades are canceled according to strictly followed rules established in the CME Globex Error Trade Policy.

Trading Session The time span marked by the start and end of the period during which the CME Globex platform is in operation. The start of the CME Globex session usually occurs in the afternoon or evening, and marks the beginning of the next trading day. For example, orders entered during Sunday's evening session are dated for and cleared on Monday. The trading day includes both the CME Globex trading session (i.e., ETH) and the trading floor open outcry session (i.e., RTH), if the CME Globex and open outcry sessions overlap for a given product.

Virtual Private Network A private data network that makes use of the Internet's public telecommunication infrastructure, maintaining privacy through the use of a tunneling protocol and security procedures.

VPN See *Virtual Private Network*.



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